



Alterations in molecular response of *Mycobacterium tuberculosis* against anti-tuberculosis drugs

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Abstract

Background Tuberculosis (TB), an infectious disease caused by *Mycobacterium tuberculosis*, has plagued humans since the early middle-ages. More than one million deaths are recorded annually due to TB, even in present times. These deaths are primarily attributed to the constant appearance of resistant TB strains. Even with the advent of new therapeutics and diagnostics techniques, tuberculosis remains challenging to control due to resistant *M. tuberculosis* strains. Aided by various molecular changes, these strains adapt to stress created by anti-tuberculosis drugs.

Materials and Methods The review thus is an overview of ongoing research in the genome and transcriptome of antibiotic-resistant TB. It explores omics-based research to identify mutation and utilization of differential gene expression.

Conclusions This study shows several mutations distinctive in the first- and second-line drug-resistant *M. tuberculosis* strains. It also explores the expressional differences of genes involved in the fundamental process of the cells and how they help in drug resistance. With the development of transcriptomics-based studies, a new insight has developed to inquire about gene expression changes in drug resistance. This information on expressional pattern changes can be utilized to design the basic platform of anti-TB treatments and therapeutic approaches. These novel insights can be instrumental in disease diagnosis and global containment of resistant TB.

Keywords *Mycobacterium tuberculosis* · Tuberculosis · Antibiotic resistance · Mutation · Differential gene expression · Antibiotic effect

Introduction

Tuberculosis (TB) is a pulmonary infection that has caused countless deaths since the Middle Ages. Evidence of the disease even dates back to as early as the Neolithic ages, which later spread worldwide through trade routes [1]. Caused by the Gram-positive bacilli, *Mycobacterium tuberculosis* (MTB), TB is persistent in 25% of the world population even in the modern age [2]. Each year, tuberculosis claims more

than 1 million lives worldwide. In 2018 alone, an estimated 10 million people fell ill with TB. Among them, 5.7 million were men, 3.2 million women, and 1.1 million children, of which 25% failed to survive [3]. Even in the presence of anti-TB drugs, the number of infected cases is increasing at an alarming rate. TB has increased more aggressively in developing and underdeveloped countries over the last 10 years, and drug-resistant TB is a major cause. Resistant strains are responsible for 30% of TB cases worldwide [3]. MTB strains with in vitro resistance to first-line drug isoniazid (INH) and rifampicin (RIF) accounted for 4,80,000 cases in 2015 that increased to ~5,60,000 in 2017 [4]. Multidrug-resistant TB (MDR-TB) strains that have additional resistance to ethambutol (EMB), and pyrazinamide (PYZ) is reported to cause disease in 29% of all these cases [3]. With high mortality rates (41% untreatable MDR-TB infections in 2020), MDR-TB is a global threat to public health [5] and is further increased by the deadly co-infection with HIV/AIDS [6].

The emergence of extensively drug-resistant TB (XDR-TB) and totally drug-resistant TB (TDR-TB), which are

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resistant to fluoroquinolones (FQ) and aminoglycosides (AMN) in addition to INH, RIF, PYZ, and EMB, has intensified the crisis. 75% of patients infected with MDR-TB were found untreatable with previously used anti-TB drugs [7]. Even though new drugs are available, limited access along with the inability to construct effective regimes and high prices have contributed to the ongoing poor outcomes in drug-resistant TB, including treatment failures and mortality [7].

Differentiation of sensitive MTB from drug-resistant MTB is complicated. The conventional culture-based method is the golden standard of TB diagnosis in patients [8]. This may take several weeks to months due to the slow growth rate of MTB in vitro, causing a delay in initiating appropriate treatments. Sub-optimal treatment strategies are one of the leading causes of the development of drug-resistant strains [9]. These factors underscore the importance of successfully developing molecular identification methods that can differentiate between sensitive and drug-resistant MTB. Personalized treatment regimens can be developed for every patient with a drug-resistant form of TB to prevent the ongoing resistant TB transmission and effectively control this global pandemic [10].

The principal mechanisms of drug resistance are mutations and differential expression of genes involved in various metabolic pathways [11]. Spontaneous or acquired mutations resulting in drug-target alterations are a standard mechanism involved in resistance [12]. These mutations are mainly in the form of insertions or deletions. Unlike other bacteria, resistance in MTB is not developed by horizontal gene transfer [11]. Alternative mechanisms, such as reduced drug permeability of the cell wall and aggressive efflux, are likely to be essential for conferring resistance in isolates, where no target gene mutations are identified [13]. Compared to mutational resistance, the expression of genes is much more varied and widespread in strains of MTB as the organism must adapt to exogenous stress conditions to survive [14]. Thus, differential expression of transcriptional regulators controlling gene sets that respond to environmental stimuli is an essential mechanism for survival in stress. The difference in gene expression is often associated with resistance to widely used antibiotics [15, 16].

Drug-resistant strains of MTB use more than one mechanism to avoid stress. Some strains increase their division time under stress from antibiotics. It was found that *Mycobacterium* tends to increase its cell division time under hypoxic conditions to tolerate stress better [17]. The impermeable MTB cell wall works to provide an active barrier against antibiotic entry [18]. Activation of efflux pumps is also linked to drug evasion [15]. Biofilms formation induces resistance to various stress, including antibiotics [19]. These responsive changes are controlled by multiple regulators, such as the Toxin-Antitoxin (TA) proteins [20], the

DosS-DosT system [21] and the *WhiB* protein family [22], *Esx* secretion system under the influence of the *RegX-SenX* two-component system [23]. These stress-based systems regulate the expression of genes that allow MTB to tolerate and even resist drug stress [20–22].

Resistance, tolerance, and persistence

Resistant MTB has the inherent or acquired ability to withstand bactericidal drugs at concentrations greater than or close to minimum inhibitory concentrations (MIC) that can otherwise kill these cells. However, tolerant strains can bear with drug concentrations lower than MIC [24]. The molecular changes accompanying MTB in developing tolerance against antibiotics is similar to resistance. On the other hand, persistence is a phenotypic character of MTB cells [25]. A population of cells treated with TB drugs have fewer persistent cells that can evade drug stress even at concentrations higher than MIC. These cells can rejuvenate the MTB population once the conditions become favorable [25].

Changes that accompany persistence involve cellular dormancy characterized by decreased metabolism and a reversible decrease in growth rate [26]. Baek and his coworkers have reported increased triacylglycerol (TAG) formation in persistent cells [27]. High TAG has often been correlated with the initiation of dormancy in bacterial cells [28]. Other than TAG, transcription activator (TA) proteins that are under the control of guanosine pentaphosphate (p)ppGpp can also be involved in reversible metabolic dormancy [29].

As most of the literature that distinguishes resistance, tolerance, and persistence were based on physiological observations [30], molecular mechanisms determining the three were significantly less. Due to this lack of information, we have limited the focus of this review only to those changes that were expressed in resistant MTB strains under drug stress.

Resistance development in MTB can be both pre-transcriptional and post-transcriptional; even then, pre-transcriptional mutation remains the most commonly used approach to explain resistance development in MTB. However, with the development and widespread use of gene expression assay, a considerable amount of information is generated to extend our knowledge of resistance by understanding expression changes [31]. As resistant MTB strains can develop tolerance to different drugs via target-based mutations, alternate methods to control these strains are required. In this regard, information regarding expressional changes can be necessary as a secondary target for future anti TB therapies.

This study is a comprehensive analysis of the differential expression of various genes available in the literature. It combines RT-PCR, qPCR, and Micro-array and RNA seq. analysis of transcripts. Different transcript changes for commonly used anti-TB drugs viz. INH, RIF, EMB, and the

combination of drugs were studied. Changes in gene expression for drugs explicitly used in MDR-TB like FQ (ciprofloxacin, CIP; ofloxacin, OFX); AMN (amikacin, AM; kanamycin, KAN; streptomycin, SM), as well as glycopeptide drug vancomycin (VAN), were also covered. This study aims to identify specific transcriptional patterns that will help define the mechanism underlying drug resistance and differentiate drug-resistant MTB from sensitive MTB strains.

Collection of relevant information

Text mining of literature

Text mining extracted relevant scientific datasets from published literature and classified them into inclusion/exclusion criteria. The scope of the investigation was the study of mutations and differential gene expressions produced by the use of specific first-line and second-line TB drug-like INH, RIF, EMB and SM, FQ, VAN, AMN. Mutations and differential expressions were confined to the MTB. Changes in the transcript in macrophages were not covered in this study.

Source of information

A systematic search was carried out in PUBMED and GOOGLE SCHOLAR with the following combinations of the keyword “Mycobacterium first-line drug and mutation”, “Mycobacterium and second-line drug and mutation” “Mycobacterium and Differential gene expression”, “Mycobacterium and Antibiotic stress”, “Mycobacterium and Antibiotic stress and Differential gene expression”, “Mycobacterium and differential protein expression”. The research papers and articles were included if they met the following criteria: (i) Studies were based on MTB in either lab culture or clinical samples. (ii) studies describing the mutations produced from the antibiotic stress and/or (iii) studies that focused on the differential expression of various transcripts in either laboratory strains or clinical samples when put under antibiotic stress. (iv) qRT-PCR, Microarray analysis, RNA seq. information was used to determine the differential expression of transcripts. The study was confined only to articles written in the English language. Reference lists of related papers were manually searched for relevant articles and research papers.

Extraction of information

Titles of individual articles were checked to reduce duplicate articles. After removing similar articles, abstracts were used to ascertain the importance of the paper in the present study. The abstracts with information about different first-line (INH, RIF, EMB, PYZ) and second-line (FQ, AMN,

VAN) anti-TB drugs along with mutation and/or differential expression in MTB were selected for full article study. Qualitative and quantitative analyses of the different transcripts were used for the study. Required data were extracted from the selected articles and their supplementary materials in public databases. The data thus collected were analyzed for biases using statistical methods. After eliminating biases, the remaining literature was used in this study (Fig. S1).

The protein products of mutated genes and differentially expressed proteins were obtained from the Mycobrowser database (<https://mycobrowser.epfl.ch>). Genes were classified based on their functional roles. Based on the folds change in product formation reported by different authors, the genes were further classified into upregulated and down-regulated genes based on fold change.

Analysis and visualization

Data analysis and inferences were drawn by performing statistical evaluation and visualization using R statistical software. The datasets were transformed and standardized using the “*decostand*” function from the “*vegan*” package if required before ordination the contingency tables to understand the effect of different antibiotics on specific categories of genes. It was also used to determine which gene categories were affected by different TB drugs. The average values were used for identical genes from other studies with different transcript quantification values. R was also used to visualize the various categories of genes based on their fold change in transcript under stress. The diagrams were prepared using Adobe Illustrator software. The figures summarize the information obtained via text mining of literature regarding changes in gene expression of efflux pump exporters and other genes. They present a possible mechanism of resistance based on the mutations and gene expression changes; however, they may not be considered the sole mechanisms involved.

Molecular responses against anti-tuberculosis drugs

Isoniazid causes over-expression of MFS family efflux pumps

INH is a prodrug that operates by inhibiting the formation of the actinobacterial cell wall. The cell wall plays a prominent role in pathogenesis [32] (Fig. S4). INH is converted into isonicotinic acyl radicals inside the cell by the Mycobacterial catalase-peroxidase enzyme encoded by the *katG* gene [33, 34]. The most common mechanism involved in INH resistance is the mutation of the *katG* gene, which commonly involves the substitutions S315T and T275P (Table

SI) [35]. However, *katG* mutations are inherently linked with an increased cost of fitness for MTB in the host without stress. To counterbalance, MTB can accumulate mutations in the promoter of *aphC*, which results in the restoration of detoxification activity of MTB cells and hence partially exempt the increased fitness cost [36]. Reports of mutation in *inhA* and *inhB* or their promoter region are also linked to INH resistance (Fig. 1) [37, 38]. However, unlike *katG*, the effect of *inhA* and *inhB* mutations in the overall fitness cost of MTB is unclear [36].

In terms transcriptional expression changes, susceptible strain H37Rv showed expression of 89 genes under INH stress [39]. Approximately 30% of the expression is related to the cell wall and cellular processes. It is accompanied by some conserved hypothetical genes with unknown roles (Fig. 2A). Lipid metabolism genes have also been greatly expressed. The gene *lprj* (Rv1690), which is a putative lipoprotein identified as a non-essential gene showed a substantial increase in expression (52 folds) [39, 40]. This was

followed by an uncharacterized protein (Rv2293c) with an increase of 17 folds. Some of the genes were downregulated (Fig. 2A). Downregulation of Rv1072, a transmembrane protein, and the NO_3^- transporter *narK2* (Rv1737c) genes was prominent. This indicates tolerance developed against oxidative stress (Table SII) [40].

When the strains resistant to INH (viz. mono-resistant strains) were induced with similar stress, their gene expression profile showed a substantial difference (Fig. 2B). All the genes expressed in these strains were involved in cell walls and cellular process pathways [41]. Nearly 60% were different forms of efflux pump genes. 50% of these efflux pumps were transporters of the MFS family. Notable among these were Rv2333c, an MFS family SM and tetracycline pump, *drdA* (Rv2936), an ABC family efflux pump responsible for doxorubicin resistance, and Rv2459 and RND type efflux pump (Table SIII) [41]. These expressions were often coupled with a mutation in *rpoB* genes [42]. However, overexpression of the pump alone has shown resistance to RIF

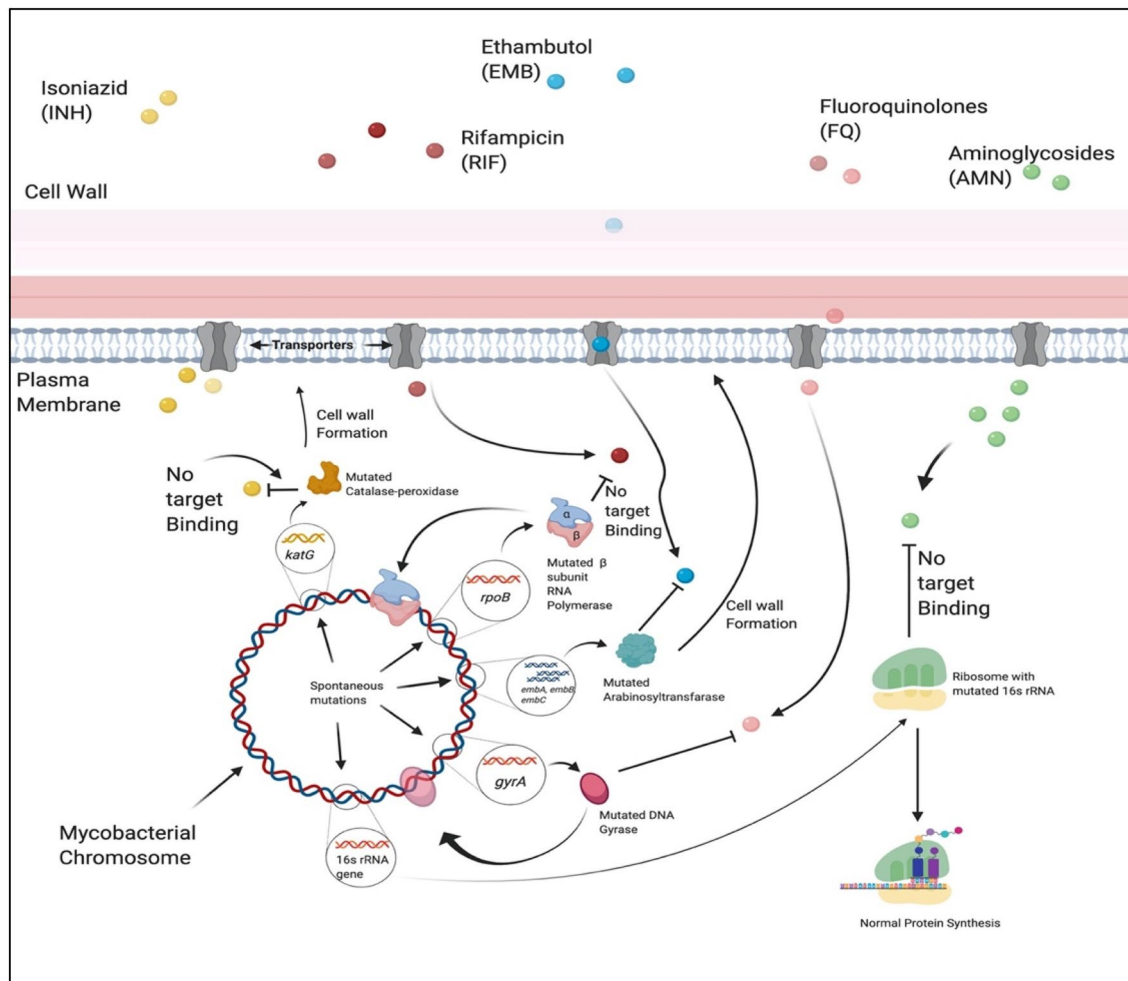


Fig. 1 Mutations specific to different genes in the genome of MTB can develop resistance against both the first line and second line of anti TB drugs. Spontaneous mutations in the MTB genome result in mutated proteins with conformational changes in the target anti TB drug binding site

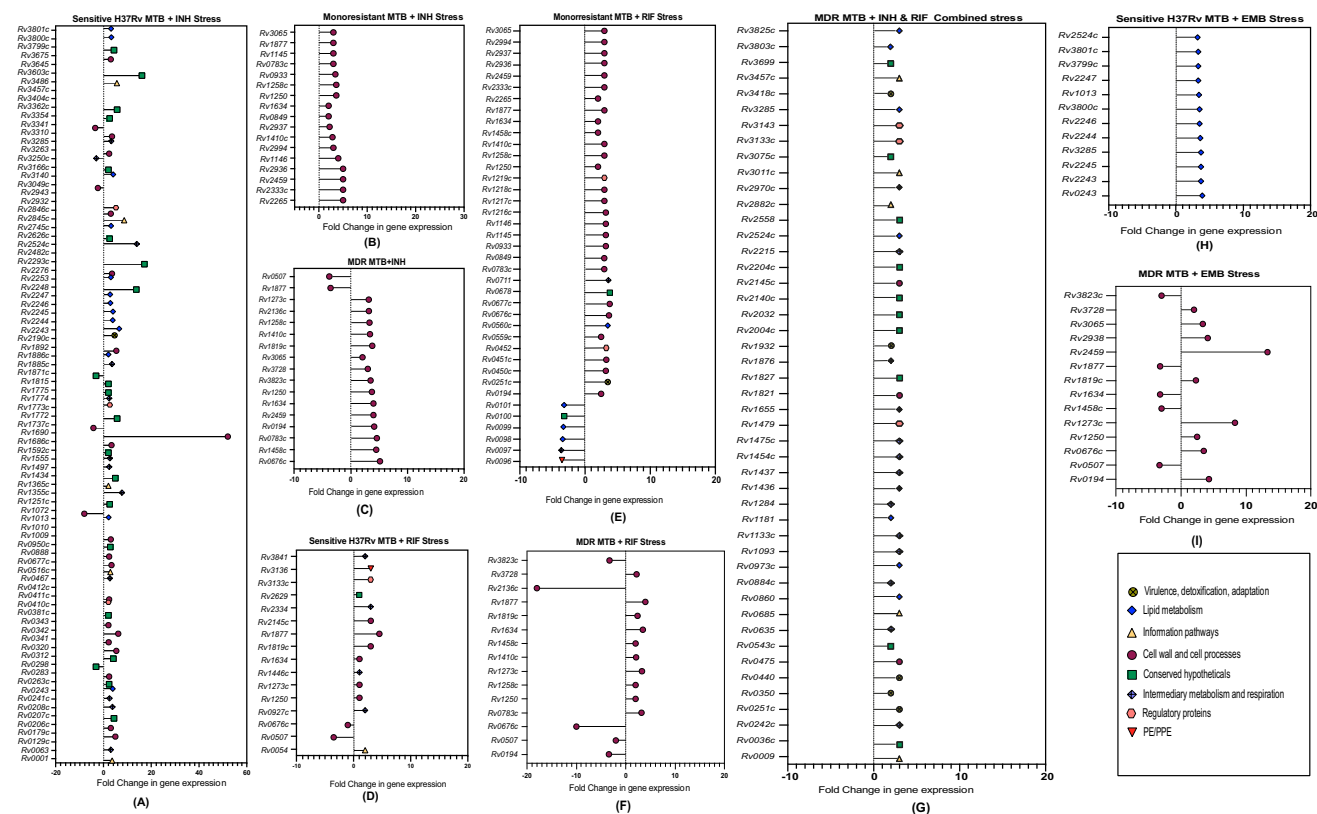


Fig. 2 Differential expression of different genes under stress from the first line of TB drugs. Here different shapes and colours of the points indicate different pathways that are differentially expressed under drug stress. The zero line distinguishes the upregulated genes (on the right of zero line) from the downregulated ones (on the left of zero line); the further right from the zero line, the more is the gene upregulated and vice versa. Here **A** represents all the differentially expressed genes in sensitive H37Rv strain of MTB under INH stress; **B** shows the differentially expressed genes in monoresistant MTB

and INH [42]. It is also worth mentioning that, unlike the sensitive strains, none of the genes has been down-regulated in INH-resistant strains.

The MDR strains are inherently resistant to INH and other commonly used TB drugs like RIF and EMB. It uses a mechanism similar to the mono-resistant strains to tolerate INH stress [43]. 17 genes all related to the cell wall and cellular processes showed changes in expression (Fig. 2C). Approximately 80% of these genes were efflux pumps, of which ~60% were the MFS family of transporters (Table SIV). Hence, the MFS transporters play a significant role in INH tolerance. Considerable expression was observed for *mmpL5* (Rv0676c), a siderophore exporter, and *emrB* (Rv0783c), a multidrug transporter in RNA-seq. analysis of MDR strains [44, 45]. Two genes were downregulated, which include *mmpL2* (Rv0507) and Rv1877. It is to be noted that *mmpL2* has no impact on cell growth; however,

under INH stress while **C** shows similar results in MDR-MTB under INH stress; **D** represents the up and downregulated genes in sensitive H37Rv under RIF stress while **E** is differentially transcribed monoresistant MTB genes under stress from RIF, **F** is the genes of similar category differentially expressed in MDR strains. The combined stress of INH and RIF express different genes shown in **G**. EMB stress **H** and **I** represent the differentially expressed genes for sensitive H37Rv and MDR strains

disruption of the RND type transporter Rv1877 facilitates the growth of MTB [39].

The prominent involvement of MFS transporters in INH tolerance is evident from the gene expression profile. Possible participation of efflux pumps in pumping out drugs outside the cell can be considered (Fig. 4A). Several studies with efflux pump inhibitors have inferred the lowering of MIC in clinically resistant strains and thus can be a viable approach for treating MDR-TB strains. Targeting the MFS family efflux pump with specific inhibitors may also benefit the treatment of MDR-TB strains.

ABC transporters give rifampicin tolerance

Unlike INH, RIF target the RNA synthesis process by targeting β subunit of DNA dependent RNA polymerase [46]. Hence, RIF-resistant TB (RR-TB) caused by RIF resistant strains as well as MDR-strains has a mutation in the *rpoB*

gene [47, 48] (Fig. 1). Substitutions H526D and S531L are commonly observed mutations in *rpoB*. However, substitutions L452P and D435G in *rpoB* are also reported in RNA-Seq. analysis [35]. While the latter does not affect the overall fitness cost of MTB, the former can increase the fitness cost of MTB in the host. RR-MTB can ease their fitness cost by accumulating mutations in *rpoA* and *rpoC* genes [49]. H256A mutations in *rpoB* are further compensated by mutations in the *gyrA* gene. Due to the lack of information available, the effect of DNA gyrase (from mutated *gyrA*) on RNA polymerase β subunit (from mutated *rpoB*) could not be confirmed. However, it could be speculated that the mutated DNA gyrase may be better adapted for the conditions devoid of RIF stress hence having decreased fitness cost.

When RIF sensitive strains, devoid of *rpoB* gene mutation, are targeted with RIF stress, a total of 19 genes were expressed differentially [18]. Like INH stress in RR resistant and, MDR strains a majority of the genes (~60%) belonged to the cell wall and cellular processes, and the majority were efflux pumps (Fig. 2D). Interestingly the gene Rv1877, which was down-regulated in INH stress, was highly upregulated in RIF stress. Vitamin B₁₂ transporter *bacA* (Rv1819c) and cell wall synthesis protein *wag31* (Rv2145c) were also upregulated. *bacA* is a non-essential gene of MTB and is responsible for enhanced in vitro growth. *wag31*, on the other hand, is an essential gene (Table SV). It maintains *pbpB*, a penicillin-binding protein responsible for in vitro growth under oxidative stress. The *mmpL* efflux pumps *mmpL2* (Rv0507), *mmpL5* (Rv0676c) and *mmpL8* (Rv3823c) were downregulated, but their changes were inconsiderable. However, increased activity of *mmpL8* was also observed in acid stress conditions [50].

The RR-MTB strains that were resistant to RIF have also been studied. When RIF induced these strains, 37 genes showed changes in expression [51]. Of these, about 70% belonged to the cell wall, and the cellular process (Fig. 2E) and 40% of these genes were efflux pumps. The majority among these were benzoquinone methyltransferase (Rv0560c), ABC-type doxorubicin transporter *drxA* (Rv2936) (Table SVI) [51]. The genes involved in lipid metabolism, however, were downregulated. The reason for this downregulation is unclear. Some authors have linked changes in lipid synthesis with the initiation of biofilm-like structures [51].

RIF stress on MDR-TB showed overall expression comparable to that of INH. 17 genes, all parts of the cell wall, and cellular processes were either upregulated or downregulated. 76% of these genes were some types of efflux pump (Table SVII). The ABC family transporter occupied the majority as 61% of the pumps were of the ABC family. This indicates that the MDR strains actively pump out RIF from inside the cells for better adaptation to stress [52]. The expression of upregulated genes had more minor variations among them.

However, Rv2136c, the gene for a conserved transmembrane protein, was highly downregulated (18 folds). The gene is vital in tolerating acid stress and is required for mycobacterial growth inside macrophages [53]. The reason for the downregulation of this gene is not understood.

ABC transporters were responsible for the active transport of compounds (Fig. 4A). Researchers have found the involvement of ABC transporters in the efflux of drugs across the membrane in other infectious bacteria as well [52]. Understanding the specific ABC transporters, target-based drugs could be designed to better counter-resistance in MTB [52].

MTB use MFS and ABC family pumps to resist Ethambutol stress

EMB functions by hindering cell wall synthesis. EMB disrupts arabinogalactan synthesis by inhibiting the enzyme arabinose transferase [54]. MTB resistance to EMB is mostly by spontaneous genetic mutations (Table SI). RNA-Seq. analysis have shown mutations in the genes mainly *embA*, *embB*, *embC*, *embR* and *ubiA* in MDR-MTB [35, 55]. These mutations result in mutated arabinosyl transferase enzyme with loss of EMB binding sites (Fig. 1). Hence, even in the presence of EMB, cell wall development is not halted [56, 57]. But the commonly observed substitution M306V reported in *embC* has fitness reducing effects on MTB which is disadvantageous to the in vivo MTB under no EMB stress for long durations. To recuperate, some of the cells have developed a mutation in the arabinofuranosyl transferase (*aftA*, Rv3792). This enzyme catalyzes the addition of the first critical arabinofuranosyl residue to the galactan domain of the cell wall from the sugar donor β -D-arabinofuranosyl-1-monophosphoryldecaprenol, thereby initiating the galactan for subsequent synthesis by arabinofuranosyl transferases. This mutation decreases fitness cost allowing the MTB to persist in the host for longer durations (Fig. 1).

While considering the gene expression profile in the sensitive strains under EMB stress, 24 genes were expressed. 44% of these genes were linked to the cell wall, and cellular processes and 48% were lipid metabolism genes [18]. Efflux pump genes were present in the majority. Both MFS and ABC family transporters were upregulated in equal terms (Table SVIII) [58]. An ABC transporter drug transporter (Rv1273c) increased the number of transcripts, followed by another multidrug transporter, Rv0194 [59]. Among the downregulated transporters more prominent were the *mmpL2* and *mmpL8* [58], a siderophore and sulpholipid exporter, respectively [60]. All of the genes of lipid metabolism were upregulated; however, the expression is not different for unstressed MTB (Fig. 2H). Data on strains resistant

only to EMB is unavailable. Hence, no conclusion regarding such strains could be drawn.

MDR-TB under EMB stress had a gene expression pattern similar to MDR-TB under INH stress. This may be because both the anti-TB drugs work by inhibiting the cell wall synthesis; however, they differ in their target [33]. Like INH, all the genes differentially expressed in EMB stress belonged to the cell wall and cellular processes (Fig. 2I, Table SIX) [54]. Most of them (~90%) were efflux pumps, of which 41% were MFS family while 50% were ABC family [60]. The MFS pump *jefA* (Rv2459) was exponentially upregulated, and it is actively involved in pumping out INH and EMB in resistant strains [56]. Studies with efflux pump inhibitors have shown that *jefA* is more active in pumping out EMB than INH [56]. The MFS pump Rv1877 was, however, downregulated. The reason for this downregulation is not understood [61]. Most of the ABC family transporters were upregulated except for Rv1458c. This drug transporter was downregulated by 3 folds. Studies have shown that change in expression of this gene can be associated with resistance to any of INH, RIF, EMB or all of them [62].

Like INH and RIF, MFS and ABC family of transporters were involved in the efflux of EMB as well (Fig. 2 H-I). It is, however, worth mentioning that further research is needed to establish a relation between EMB and the said families of drug transporters.

MDR strains evade combined stress of Isoniazid and Rifampicin by changing its metabolism

The gene expression profile of MDR strains under the combined stress of RIF and INH were readily distinguishable from similar strains treated with only one anti-TB drug. This is attributed to the fact that both INH and RIF have entirely different mechanisms of action [33]. Strains with combined mutations in *rpoB* and *katG* can readily resist this combination of drugs [44] (Fig. 1). MDR strains showing resistance to this combination of stress is not uncommon [52]. MTB strains can acquire such mutations via spontaneous mutations.

As is mentioned previously, *katG* and *rpoB* mutations are responsible for reduced fitness in MTB, a combined effect of non-synonymous mutations in both *aphC*, *rpoA*, *rpoC* and *gyrA* play a crucial role in regulating this fitness cost even when there is combined stress of both INH and RIF [63].

Consequently, mutations provide only half of the picture. MDR strains have changes in their gene expression profile as well. The expression of genes differed from the treatments with single drugs. It was seen that of 47 genes that were differentially expressed, 30% belonged to the metabolism and respiration pathways (Fig. 2G, Table SX). Approximately 20% were conserved genes with unknown functions [60]. Upregulation was prominent in genes of the information

pathway and the respiration and metabolism pathway [56]. No downregulation was observed in any of the expressed genes. This indicates that metabolic changes in respiration and lipid synthesis, information pathways interact in manners that help them tolerate oxidative stress conditions [64]. Combined INH and RIF present oxidative stress; hence it can be speculated that overexpression genes of protein folding (Rv0440, *groEL2*) and cell wall synthesis (Rv2145c, *wag31*) are ways to tolerate oxidative stress [61]. It is also worth noting that studies targeting the other combination of INH, RIF and EMB were unavailable. Hence, changes in gene expression under other combined stresses were not addressed.

MTB resist fluoroquinolones by efflux activity and changes in information pathway

The essential target of FQ in MTB is DNA gyrase (a type II topoisomerase, comprised of two A and two B subunits) [65]. The mutations affecting the FQ resistance was observed in a small region of *gyrA*, known as quinolone resistance-determining region (QRDR), and somehow in a negligible amount in *gyrB* was found as the standard mechanism of FQ resistance in (Fig. 1) [66]. However, unlike the mutations responsible for resistance in first-line drugs, *gyrA* and *gyrB* mutation's effect on the overall fitness cost in host incorporated MTB is unclear due to lack of study.

For MDR-TB treatment, the WHO list added a synthetic FQ drug OFX which inhibits mycobacterial cell replication and inhibits enzymes responsible for DNA separation [61]. Mutations in *gyrA* were linked to OFX resistance in XDR-TB strains [67] (Fig. 1).

OFX was found effective when used for MDR-TB isolates. Uses in vivo caused upregulation of two ABC transporter proteins from the cell wall and cell process categories. Rv2477, an ATP dependent macrolide transporter that is an essential gene for mycobacterial growth in vivo [39]. Their role in OFX resistance is unclear [68]. It can be predicted that Rv2477 may actively pump out OFX, not allowing OFX to accumulate inside the cell. The conserved integral membrane protein Rv2209, a non-essential gene, is also overexpressed [39]. However, authors have reported that disruption of this gene had resulted in better in vivo growth in slow-growing MTB (Fig. 3A, Table SXI). More study is required to understand the significance of overexpression of transmembrane protein in MDR isolates. It should be mentioned that, in drug-sensitive strains, the role of multiple genes such as transcription elongation factor, heat shock protein, thioredoxin, a certain unique class of kinase, ligase, isomerase response to OFX stress is still uncertain. No common genes could be classified for MDR and sensitive strains (Fig. 3A).

Ciprofloxacin (CIP) is commonly used for resistant pulmonary TB treatment [61]. CIP is a broad-spectrum

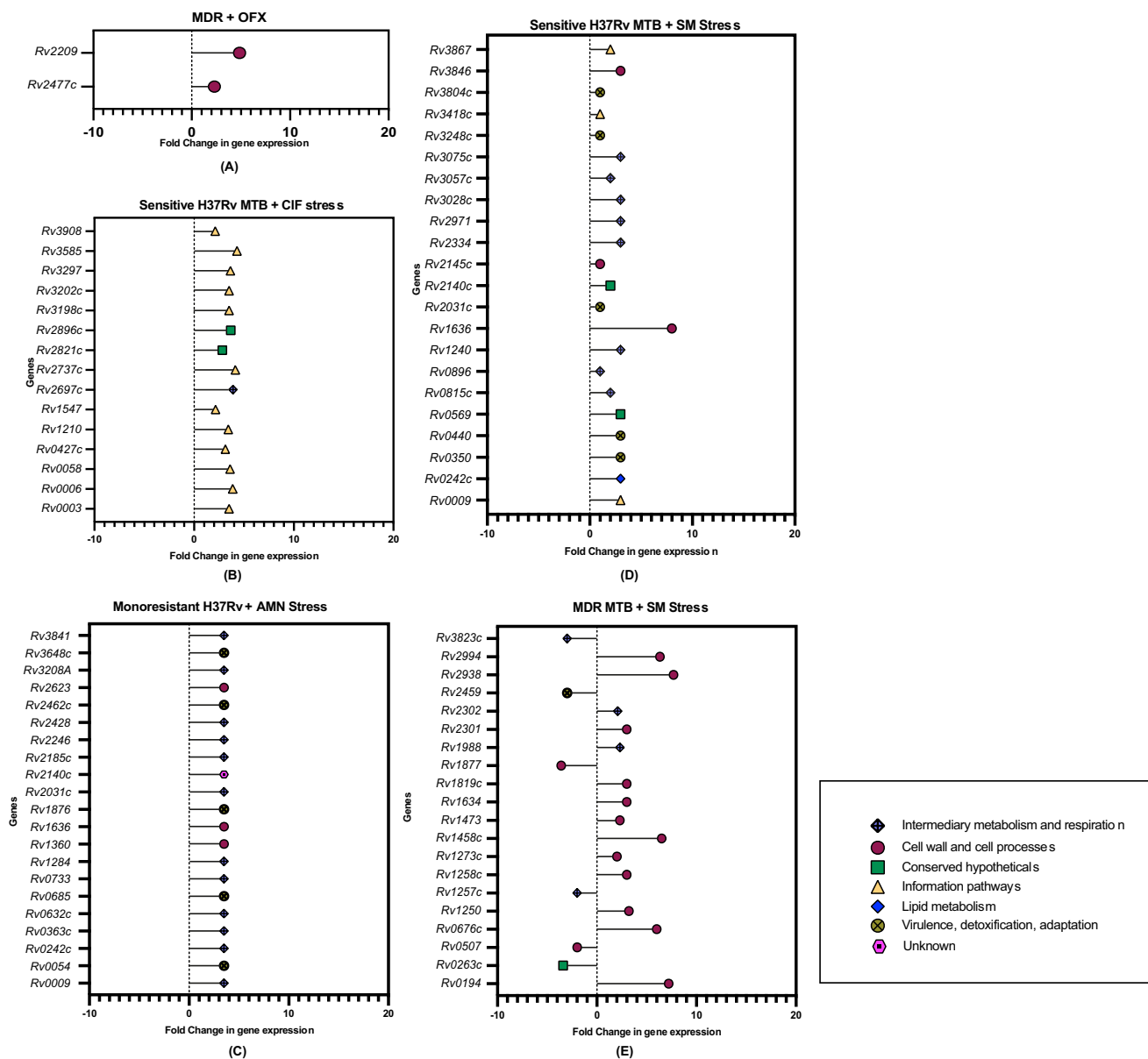


Fig. 3 Differential expression of different genes under stress from the second line of TB drugs. Here different shapes and colours of the points indicate different pathways that are differentially expressed under drug stress. The zero line distinguishes the upregulated genes (on the right of zero line) from the downregulated ones (on the left of zero line); the further the point is to the right from the zero line,

the more is the gene upregulated and vice versa. Here (A) shows differentially expressed genes for MDR strain in OFX stress (B) is varied expression sensitive H37Rv genes in CIF stress while (C) is the H37Rv under AMN stress (D) and (E) shows the differentially expressed genes for SM stress in sensitive H37Rv and MDR-TB strain

antibiotic that can be used against both Grams positive and Gram-negative bacteria. It functions by inhibiting DNA gyrase and type II topoisomerase, topoisomerase IV [61] required to divide bacterial DNA, thus inhibiting cell division. Mutations in the *gyrA* gene [69] were directly linked to CIP resistance in MTB [70](Fig. 1).

CIP stress in MTB also induced 16 specific genes, and some of these were *tagA* (Rv1210), *recF* (Rv0003), *uvrD2* (Rv3198c), *nei* (Rv3297). These are essential genes

invariably involved in DNA repair and maintenance [39]. These genes' expression in drug-sensitive strains indicates changes in the information pathway to tolerate CIP stress [67]. They were involved in various information pathways such as DNA defense, repair, replication and recombination with increased expression level (Fig. 3B). Interestingly, upregulation was also observed in the gene involved in the SOS response indicating an accelerated *dnaE2* (a DNA repair gene) activation (Table SXII).

From the gene expression pattern, it was seen that FQ increases genes of DNA repair and maintenance that can be necessary for keeping the DNA defected by improper gyrase activity (Figure 4B) [65, 68]. Thus, the MDR strains can tolerate FQ concentrations equal to greater than MIC, indicating the development of XDR characters.

Aminoglycosides induce tolerance by changing metabolism in MDR

AMN family of the drug includes major injectable AK, KAN and capreomycin (CAP) predominantly inhibits the protein synthesis in MTB, and most importantly, the resistance to these drugs was linked with changes in the 16S rRNA (*rrs*) gene [71]. Besides, A1401G and C1402T mutations were associated with CAP and KAN resistance, respectively [35]. Mutation in the *rrs* gene is linked to high-level AK, KAN resistance, and low CAP resistance (Fig. 1) [72, 73].

Cytosolic proteome analysis of AK and KAN resistant MTB revealed several genes responsible for intermediary metabolism were expressed differentially [74]. In this study, 60% of overexpressed genes were found to be involved in homeostasis/biosynthesis of fatty acid, protein folding, the protein responsible for changing the cell permeability (Fig. 4B), which helped in the survival of the bacteria under the effect of AMN stress. Among them, *bfrA* (Rv1876), a protein that is essential for iron storage, was assumed to play a role in the regulation of the metabolism of iron and its contribution to resistance (Fig. 3C, Table SXIII). It was speculated that the disruption of *bfrA* provides a growth advantage to some MTB strains [39]. Other hypothetical proteins like Rv3224, a non-essential iron regulator [39] and the secretion protein *espH* (Rv3867) were also expressed. Further validation is needed for their role to advance TB therapy along with suitable diagnostics [74].

Similarly, other studies of AK and KAN drug-related stress had shown overexpression of the secretory protein *espH* (Rv3867), thiol peroxidase gene *tpx* (Rv1932), *groES* (Rv3418c), bacterioferritin gene *bfrA* (Rv1876) [75]. Their over-expression in AK and KAN resistance calls for a detailed study of the biological pathways responsible for their differential regulation. These overexpressed genes have importance in new anti-TB drug development to enhance the understanding of XDR-TB.

AMN disrupts protein formation, and changes in gene expression of genes related to protein folding and maintenance indicate the cell's ability to adapt to this stress [74]. Overexpression of *groES* and similar genes is indicative of changes in protein formation and function [76]. Other over-expressed genes like *tpx* [77], *espH*, *bfrA* [78] may help the

bacterium tolerate stress by inducing changes in metabolic pathways (Fig. 4B).

Streptomycin stress causes MTB cells to overexpress enzymes of lipid synthesis

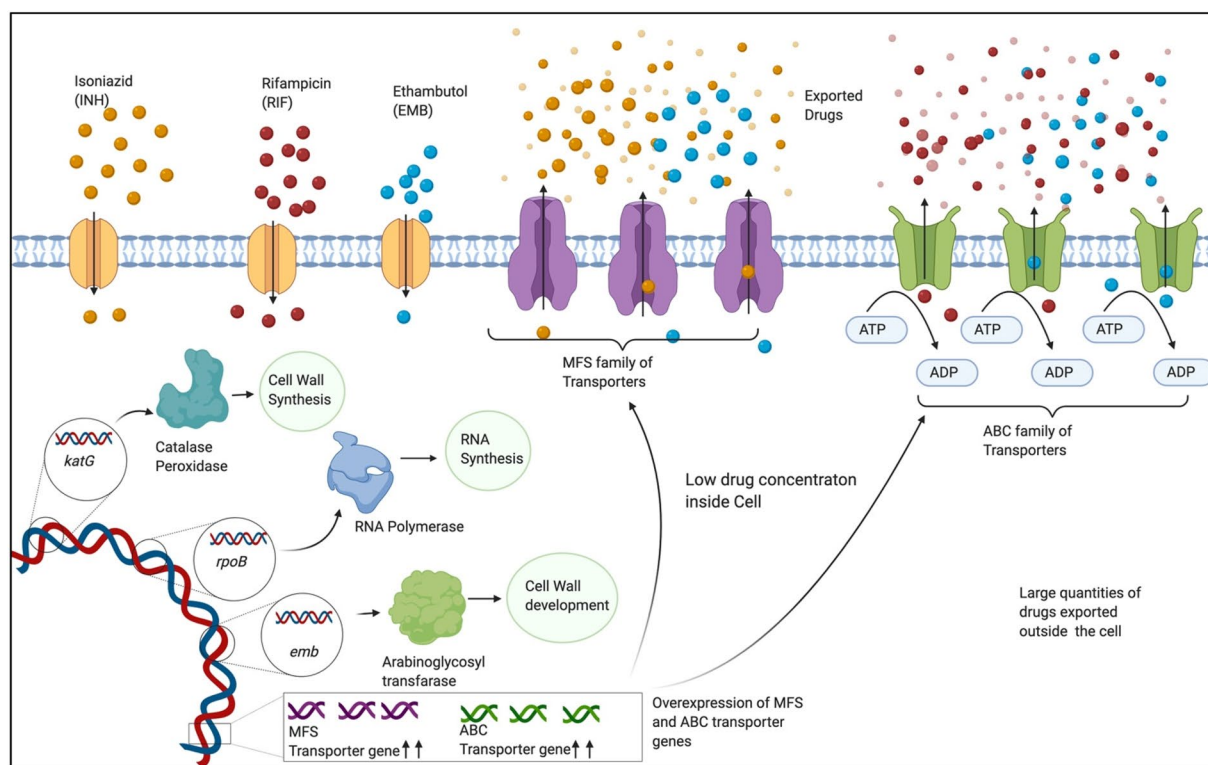
SM is an aminocyclitol glycoside drug that binds to the 16S rRNA involved in translation proofreading, hindering protein synthesis. SM resistance occurs due to mutations associated with the genes encoding 16S rRNA (*rrs*) [9] and S12 (*rpsL*), which are ribosomal proteins [79–81]. Various studies suggested that the alterations in the 16S rRNA structure can interrupt interactions between 16S rRNA and SM, resulting in SM resistance [82] (Fig. 1).

In vitro study on the sensitive strain with SM, stress caused overexpression of a series of 10 transporter family genes under cell wall and cell process [83] these include genes like the ABC family efflux pump (Rv0194), the transmembrane antibiotic exporters (Rv1458c, Rv1250, Rv1634, Rv1273c), the RND transporter: *mmpL5* (Rv0676c) [58] and MFS transporter *mmpL2* (Rv0507) [58], *bacA* (Rv1819c), a conserved transmembrane protein (Rv1877), and *mmpL8* (Rv3823c) [83] (Table SIV). Among upregulated ABC transporter family, multidrug efflux ATP binding permease protein Rv0194 showed the highest change and downregulated Sulfolipid-1 exporter protein Rv3823c exhibited down-regulation [83] (Fig. 3D).

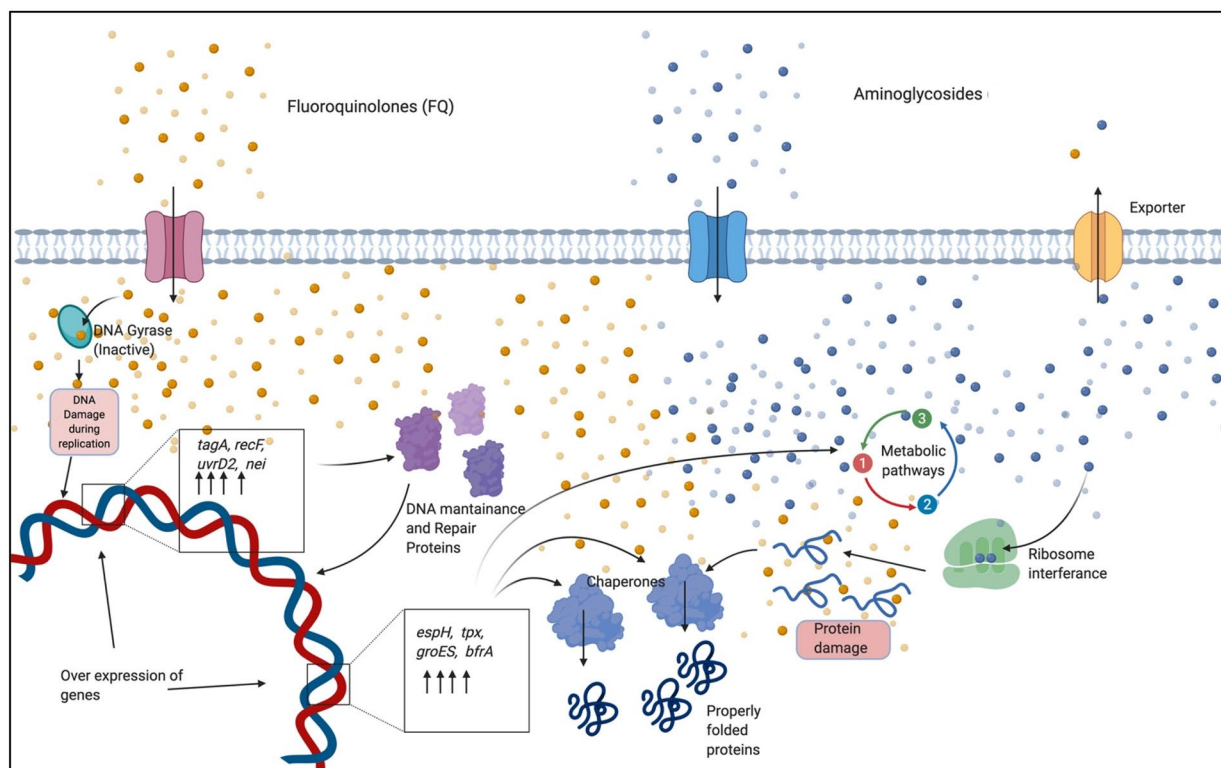
Analysis of MDR-TB strains of MTB showed overexpression of specific essential and non-essential genes [84]. Overexpression was observed in the presence of SM, for cell wall and cell process gene (Fig. 3E, Table SV) [74]. These include ABC family efflux pump Rv2688 & Rv2938 with potential roles in drug efflux. Two conserved integral membrane proteins of MFS transporter family Rv2994 and Rv2459 for MDR isolates, where Rv2938 carries the highest mean fold change of expression and Rv2688 with the lowest overall expression as compared to other genes [66]. Several ribosomal genes and some other proteins like thiol peroxidase, acyl carrier protein dehydrogenase (Rv0824c) were found to be down-regulated in MTB clinical isolates under SM stress [42] (Fig. 3C). A detailed study is required, giving additional information on the drug, its action, and probable binding sites. The development of new therapeutics targeting the identified genes and their products can be helpful in diagnostics development.

Vancomycin can induce virulence, detoxification, and adaptation pathways in MTB strains

Vancomycin (VAN), a class of glycopeptide antibiotics, is used for various bacterial infections. In some instances, it has been used against MTB. VAN inhibits cell wall synthesis by targeting D-alanine present between two layers of NAM/



(A)



(B)

Fig. 4 Diagrams representing the potential role of differentially expressed genes in the process of resistance development. **A** shows the mechanism by which INH, RIF, and EMB cause inhibit or kill resistant H37Rv strains. **B** Shows the monoresistant or MDR MTB strains where the overexpression of efflux pump genes of MFS and ABC type in first-line drug stress which either passively or actively transports the first-line antibiotics accumulated inside the cells allowing the target proteins to escape the effect of drugs. **C** Shows the possible action of differentially expressed genes for second-line drugs FQ and AMN and how this expression may help MDR-MTB strains resist accumulated drugs

NAG layers in gram-positive bacteria [85]. MTB resist VAN by modifying the D-ala-D-ala sequence inhibiting the binding of VAN [57].

Genes such as Rv2623 (*uspA*-like protein), Rv0116c (Probable L, D-transpeptidase LdtA), PE20-PPE31, (*serBI*), *pspA* were found to be overexpressed in RNA seq. experiments. These genes were involved in the toxin-antitoxin system [85]. Toxin proteins affect metabolism by inhibiting the replication or translation of the genes involved. The antitoxin proteins help neutralize the toxins' inhibitory activity, hence providing a necessary halt to mycobacterial metabolism [20]. TA modules, therefore, provide strict control over metabolism, thus allowing safe entry or exit from persistence [25, 27, 86–88].

Also, σ E factor confers basal resistance to VAN [60]. This is a significant study highlighting the identification of various mentioned genes whose protein products could protect from the stress on the cell wall that induces resistance to multiple drugs of TB. These proteins may be potential putative targets for TB diagnostics and mostly VAN resistant TB prediction, which needs more in-depth research (Fig. 3C).

Compensatory mutations and their role in maintaining resistance

Compensatory mutations are a crucial but little-understood feature of biological evolution with far-reaching consequences. Compensatory mutations are often utilized by bacterial cells to decrease the fitness cost induced by a mutation that is important to survive under stress [89]. Likewise, mutations conferring resistance in MTB are often associated with increased fitness costs under unstressed conditions. This increase in fitness cost obstructs the reproduction and dissemination of MTB. To recuperate, MTB, during its evolution in the presence of antibiotic stress, has accumulated mutations in genes that in most cases have no impact on the fitness cost but can compensate for the increased fitness cost in resistant MTB strains [90].

As is observed in INH stress, *aphC* mutation can make up for the increased fitness cost from mutated *katG*. *aphC* (Rv2427) produces Alkyl hydroperoxidase C and is under the regulation of *ahpC-oxvR* regulator (Rv2428). The

reduction of hydrogen peroxide and organic hydroperoxides to water and alcohols is catalyzed by alkyl hydroperoxidase C, a thiol-specific peroxidase. Detoxifying peroxides that may be produced as the result of mutated catalase-peroxidase in MDR-MTB aids in cell defense against oxidative stress. It is the primary scavenger of hydrogen peroxides. INH action requires oxidative activation by catalase-peroxidase produced by *katG* in MTB. Interestingly, *aphC* overexpression was observed in AMN stress (Fig. 3C).

Similarly, in RIF resistant strains, mutations accumulated in the *gyrA* gene may promote RNA polymerase I activity to form the pre-initiation complex in addition to being more adapted for the mutated *rpoB* [91]. Likewise, in EMB, stress mutated *ahpA* helps retreat the cells from the increased fitness cost. As the occurrence of compensatory mutations in strains resistant to FQs and AMNs antibiotics is yet unknown, the scope of this study was confined to the first-line drug mutations that have been identified.

Compensatory mutation can't produce resistant traits on its own. However, they allow the low-fitness mutant strain to compete with the sensitive wild-type strains in an environment free of antibiotic stress. These mutations profoundly impact the emergence of resistance [52]. A nonfit resistant strain may undergo more mutations before going extinct until it has regained enough fitness to compete with the sensitive strain. As a result, more profound knowledge of compensatory changes and how they offset the fitness cost can be crucial in combating the growing number of resistant MTB strains.

Differential Expression analysis

Dimension reduction approach of the principal component analysis (PCA) of sensitive strains has been carried out against various antibiotics. The PCA results illustrated variations in the cell wall and cell process adjacent to EMB and RIF, and changes in lipid metabolism were mostly clustered around the INH axis. This indicated a cause-and-effect relationship between the antibiotic and these gene categories. The expression difference in genes for AMN and SM indicate a functional polarity between the two antibiotics. VAN, CIF, and OFX had similar expression variations clustered around their axis. This can be explained by the common underlying mechanism by which the MTB cells resist the drug effect (Figure S2 A).

Unlike sensitive strains, the resistant strains had more prominent variations for EMB, INH, and SM. This can indicate changes in a system well adapted to these stressors. On the other hand, RIF clustered away from INH, EMB, and SM; thus, hinting at an entirely different mechanism to handle this kind of stress (Figure S2-B). A combination of INH and RIF (as depicted in the plot by "combination") had a diverse expression of 4 different categories of genes. It is,

however, noteworthy that these categories of genes lacked expressional diversity when treated with INH or RIF alone. This may hint towards a different resistance mechanism for the combined and the single drug stress indicating. The variations in the expression of OFX and CIP were similar, but unlike sensitive strain, the changes were more distinct for information pathway genes.

The results provide an understanding of possible combinations of drugs that may be effective in treating MDR-TB. As seen from the PCA analysis, combinations like INH with EMB and SM will be less effective in treating resistant strains as they can be exported out by the expression of different families of efflux pumps. RIF, due to its functional dissimilarity to INH, EMB, and SM, has been used in combination, but with the increased development of RR-MTB strains, even these combinations were unfavorable (Figure S2 A-B).

It can be predicted that a new combination of drugs like RIF and CIP or INH and OFX may be presented as a viable alternative to commonly used combinations due to their functional differences. However, a lack of studies has limited the understanding of the changes in the bacterial cell under such combinations. This provides a viable approach for future studies regarding differential expression in MTB.

Genes with potentials for diagnosis of MDR-TB

Conserved integral membrane transport proteins *jefA* (Rv2459) [92], Rv1634 [93], Rv1250, Rv1877, *mmr* (Rv3065), *mmpL2* (Rv0507), *mmpL5* (Rv0676c), transmembrane multidrug efflux pump (Rv0194) [94] were expressed under INH, RIF and EMB stress (Figure S3A). These genes were identified as non-essential. However, gene knockout experiments have shown that disruption of any of these genes leads to a faster growth rate in MTB. The gene, as mentioned above, was reported in both single drug and multidrug-resistant strains of MTB. Interestingly, *jefA* also shows resistance against INH, RIF, and SM [60, 70, 95]. The drug transporter Rv3728 of MFS category, antibiotic-transport ATP-binding protein ABC transporter Rv1456c, the ABC family *bacA* (Rv1819c) drug transporter may also act as potential targets for diagnosis. Disruption of the *bacA* gene results in a growth defect of H37Rv [39]. MTB requires the drug-transport transmembrane ATP-binding protein Rv1273c for prolonged survival in murine macrophages [96, 97]. These genes were not involved in the pathways affected by the different first-line drugs. Their role was more in attaining a dormant state [39, 59, 98].

Similarly, in the fluoroquinolones and aminoglycosides, iron-regulated universal stress protein family protein TB15.3 (Rv1636) [99] was found to be common (Figure S3B). The protein is increasingly expressed after 21 days in murine

macrophages infected with MTB. It is involved in adaptation to stress conditions and helps MTB survive in dormant conditions [100].

Limitations

The study is limited to MTB, one of the 9 *Mycobacterium* of the aggregation responsible for TB and its symptoms. Hence a general conclusion based on the above study may not hold grounds in all cases of TB. The infectious agent may show geographical variation in patients of a different region. Further, due to the limited amount of information available regarding the bacterial adaptation pattern before, during, and after the treatment period among the patients, the medication type is also excluded from the study. The authors would also like to point out that the field of omics is holding a significant gap concerning combinatorial testing of first and second-line TB drugs and their impact on gene expression patterns.

Conclusion

As discussed in this review, the use of advanced techniques to study the differential gene and protein expression profiles in sensitive and resistant isolates has provided vast information on physiology under antibiotics-associated stress. MDR and XDR-TB are a global threat, which majorly affects the economically weaker section of society. Numerous studies on mycobacterial adaptation and persistence throughout or after treatment could ensure the importance of the relationship between drug tolerance and resistance. There is a need to establish a connection between drug stress with other environmental stresses to recognize the common factors influencing mycobacterial adaptation inside the host; hence, the host–pathogen relationship can be well studied. To expand more awareness of disease therapeutics, some significant points that must be addressed to gain a novel outcome as potential drug targets are mentioned below-

- Efflux pumps play a crucial role in drug-resistant with the expulsion of various drugs. Prolonged exposure to FQ stress is needed to understand the importance of fluoroquinolone-specific transmembrane efflux pump genes as efflux pump specific targets in XDR TB cases.
- A lack of information is available in terms of complete coverage of differential gene expressions using transcriptomics and proteomics in a single study for both MTB and MDR-TB strains. Also, a particular drug from the first or second-line category was not used for all the omics-based studies, which could not clarify the impor-

tance of a differentially expressed gene for further usage in understanding the drug–target mechanism.

- The field of omics is holding a significant gap concerning combinatorial testing of TB drugs and their impact on gene expression patterns. Primarily, no report exists for second-line drug and their combination with other antibacterial compounds for efficient treatment. Also, universal gene expression results cannot be documented from limited studies on first-line drug combinations.
- Some existing studies could not provide useful information on patient sample sources, varying treatment conditions, post-treatment response, and differentially expressed transcripts numbers, which needs a vital ground to discuss further because all these aspects contributed majorly toward the losing battle against TB.
- Most of the genes from single drug stress and combinatorial stress were found with no commonness. Drug-sensitive H37Rv and MDR strains behave differently with fewer common genes between them from the pathway's perspective.

A large area of the differential gene expression in MTB species remains uncharted territory. With the advent of molecular technologies in modern information technology, the authors ascertain that the gap in information during this study can be breached in the next few years. With the necessary arsenal of information available, better drugs could be designed and disseminated among the required population, and TB as a global disease can be contained.

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Availability of data and material The data and materials used in this study are available in the public domain.

Declarations

Conflict of interest The authors declare that they have no conflict of interests.

Research involving human participants or animals This article does not contain any studies with human participants or animals performed by any of the authors.

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